



Final
MISO DPP 2018 April South Area Study
Phase II Report

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1. Executive Summary

This report presents the results of a System Impact Study performed to evaluate the interconnection of the generators in the DPP 2018 April South Area Phase II (South Area DPP 2). The study was performed under the direction of MISO and reviewed by an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies – Entergy, Cooperative Energy, CLECO, AECL, MLGW, PowerSouth, Southern Company (SOCO), SPP, and TVA.

1.1. Project List

The original interconnection requests for DPP 2018 April South Area had a total of 49 generation projects. Projects J1029, J1133, J1134, J1136, and J1167 withdrew prior to the start of DPP Phase I. Projects J961, J964, J972, J988, J1023, J1049, J1070, J1073, J1078, J1079, J1091, J1093, J1095, J1097, J1099, J1100, J1113, J1120, J1130, J1137, J1151, J1156, J1157, J1159, J1162, J1163, J1165, J1166, J1168, and J1186 withdrew during Decision Point 1. During Decision Point 1 Project J962 reduced their NRIS request from 100 MW to 0 MW and Project J1147 reduced their NRIS request from 100 MW to 0 MW. Therefore, there are 14 generation projects with a combined nameplate rating of 2,031 MW of Energy Resource Interconnection Service (ERIS) & 1,681 MW of Network Resource Interconnection Service (NRIS). The detailed list of South Area DPP 2 is shown below in Table 1, and the Phase II study was kicked off on September 4th, 2019.

Table 1: List of DPP 2018 April South Area Phase II Projects

Project	Fuel Type	Transmission Owner	County	State	Service Requested	MW	Point of Interconnection (POI)
J1007	Solar	EES	Crittenden	AR	NRIS	200	Marion - Marked Tree 161 kV Line
J1020	Solar	CE	Jasper	MS	NRIS	250	Missionary 230 kV Substation
J1030	Solar	EES	Liberty	TX	ERIS	150	Gulf Livingston - Menard 138 kV Line
J1060	Solar	EES	St. Francis	AR	NRIS	500	West Memphis - Keo 500 kV Line
J1076	Solar	EES	St. James Parish	LA	NRIS	200	Vacherie 230 kV Substation
J1125	Solar	EES	Cross	AR	NRIS	135	Wynne South - AECC Wynne E 161 kV Line
J1142	Solar	EES	St. James Parish	LA	NRIS	20	Vacherie 230 kV Substation
J1144	Solar	CE	Covington/Lamar	MS	NRIS	96	Moselle - Columbia 161 kV Line
J1146	Coal	EES	Mississippi	AR	NRIS Only	10	San Souci 500 kV Substation
J1147	Solar	EES	St. Francis	AR	ERIS	100	Moses - McCrory 161 kV Line
J1155	Solar	EES	Mississippi	AR	NRIS	150	Driver 230 kV Substation
J1158	Solar	EES	St. James Parish	LA	NRIS	70	Vacherie 230 kV Substation
J1184	Solar	EES	Washington Parish	LA	NRIS	50	Lagen Holton - Franklinton 115 kV Line
J962	Solar	EES	DeSoto	MS	ERIS	100	Twinkletown 230 kV Substation

1.2. Total Network Upgrades

The cost allocation of Network Upgrades for the projects in the DPP 2018 April South Phase II reflects responsibilities for mitigating system impacts. The total cost of Network Upgrades is listed in Table 2 below. The costs for Network Upgrades are planning-level estimates and subject to revision in the facility studies. Details pertaining to the cost allocation has been detailed in Appendix A – Cost Allocation Summary (CEII).

Table 2: Total Cost of Network Upgrades for DPP 2018 April South Phase II Projects

Project	ERIS Network Upgrades (\$)				NRIS Network Upgrades (\$)	Interconnection Facilities (\$)		Shared Network Upgrades (\$)	Total Network Upgrade Cost for Milestone Calculation (\$)	M2 (\$)	M3 (\$)	M4 (\$)
	Steady State	Stability	Short Circuit	Affected System	Deliverability	TO Network Upgrades	TO – Owned Direct Assigned					
a	b	c	d	e	f	g	h	i	j = b+c+d+f+g+i	\$4,000/MW	(10% of (j) from Phase I) – M2	(20% of (j) from Phase II) – M2 – M3
J1007	3,685,606	0	0	246,153	22,267,419	15,241,000	1,552,000	429,894	41,623,919	800,000	1,849,571	5,675,213
J1020	2,820,000	0	0	77,080,000	0	3,337,800	4,930,005	0	6,157,800	1,000,000	745,608	0
J1030	0	0	0	0	0	14,000,000	0	0	14,000,000	600,000	1,088,306	1,111,694
J1060	5,049,242	0	0	1,644,393	56,223,014	35,192,000	2,852,000	350,509	96,814,765	2,000,000	1,395,904	15,967,049
J1076	2,215,909	0	0	0	1,724,138	6,500,000	0	0	10,440,047	800,000	0	1,288,009
J1125	295,455	0	0	167,282	15,509,568	14,316,000	1,448,000	611,346	30,732,368	540,000	3,032,486	2,573,987
J1142	132,576	0	0	0	172,414	416,982	3,886,018	0	721,972	80,000	0	0
J1144	600,000	0	0	0	0	0	5,251,585	0	600,000	384,000	348,156	0
J1146	90,909	0	0	0	0	275,000	0	0	365,909	40,000	2,586	30,596
J1147	56,818	0	0	0	0	5,015,000	143,000	0	5,071,818	400,000	3,686,669	0
J1155	2,321,970	0	0	0	0	283,000	2,666,000	0	2,604,970	600,000	662,449	0
J1158	587,121	0	0	0	603,448	6,500,000	0	0	7,690,569	280,000	414,402	843,712
J1184	530,303	0	0	0	0	2,377,000	91,000	0	2,907,303	200,000	0	0
J962	6,034,091	0	0	160,000	0	5,000,000	0	0	11,034,091	400,000	1,364,893	441,925

Analyses performed demonstrate the following transmission facilities are required to reliably interconnect this group of generators to the transmission system. ERIS Network Upgrades and NRIS Network Upgrades are shown in Table 3. Shared Network Upgrades are shown in Table 4.



Table 3: ERIS & NRIS Upgrades (Planning-level cost estimates)

Network Upgrade	TO	GI projects requiring upgrade for ERIS	GI projects requiring upgrade for NRIS	Cost of solution (\$)
Danville 115 kV 20 MVar Capacitor Bank	EES	J1007, J1060, J1076, J1125, J1142, J1147, J1155, J1158, J1184, J962		2,500,000
Fernwood 115 kV two 20 MVar Capacitor Banks	EES	J1007, J1060, J1125, J1146, J1155, J962		6,000,000
Franklin 115 kV three 43.2 MVar Capacitor Banks	EES	J1007, J1060, J1155, J962		10,000,000
J1144 POI – Columbia 161 kV Uprate	CE	J1144		600,00
J1184 POI – McComb 115 kV New Line	EES		J1007, J1060, J1125	94,000,000
Missionary 161/69 kV Replace Autotransformer #1	CE	J1020		1,410,000
Missionary 161/69 kV Replace Autotransformer #2	CE	J1020		1,410,000
Mt Olive 230 kV 20 MVar Capacitor Bank	EES	J1007, J1060, J1076, J1125, J1142, J1147, J1155, J1158, J1184, J962		2,500,000
Tezcuco - Frisco 230 kV #1 & #2 Re-conductor ¹	EES		J1076, J1142, J1158	2,500,000

Table 4: Shared Network Upgrades (Planning-level cost estimates)

Shared Network Upgrade	TO	Higher queued projects associated with SNU	Study projects associated with SNU	Cost of solution (\$)
Kuhn Road – Ebony 161 kV Rebuild	EES	J934	J1007, J1060, J1125	4,900,000

Note:

- 1) Details pertaining to upgrades, costs, and the execution plan for interconnection of the generating facility at the POI will be documented in the Facility Study for Interconnecting Generator.
- 2) Facilities that have been included as base case assumptions and the level of interconnection service that would be conditional upon these facilities being in service will be documented in the GIA (Generator Interconnection Agreement) for each respective GI request successfully achieving GIA execution.
- 3) Analysis performed shows there are five projects for Shared Network Upgrade cost allocation.
- 4) A detailed summary of the estimated cost of Network Upgrades on a per project basis can be found in Appendix B – Network Upgrade Table Per Project (CEII).

¹ This Network Upgrade is an Appendix B project in MTEP19 (Project ID 15605), therefore, as stated in section 6.1.1.1 of [BPM 015](#) if this project is approved by the Board within (1) calendar year of the GIA execution or execution of an amendment thereof, then the IC's will not be responsible for the cost of the project.

2. FERC Order 827 Compliance Review

The Final Rule of FERC Order 827 “Reactive Power Requirements for Non-Synchronous Generation”, which was issued June 16, 2016, states that “Under this Final Rule, newly interconnecting non-synchronous generators that have not yet executed a Facilities Study Agreement as of the effective date of this Final Rule will be required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.” As such, this Final Rule applies to all wind and solar projects included in the DPP 2018 April South study cycle.

In this study, the power factor at the high-side of the generator substation for each inverter-based project was calculated and reviewed. The study method is to set Qgen of each study project at its Qmax, solve the case, then record the P and Q injection on the high side of the generator substation to calculate the lagging power factor (injecting VAR to the system). The same process is then repeated by setting Qgen at Qmin to calculate the leading power factor (absorbing VAR from the system).

The results show that all projects meet the requirement to maintain 0.95 leading and lagging power factor, as shown below in Table 5.

Table 5: FERC Order 827 Review Results

Project	Pmax (MW)	Reactive Power Capability (MVar)	Proposed VAR Compensation	VAR Injection			VAR Absorption			Meet FERC Order 827 Requirement ?	Add'l VAR Needed (MVar)
				P (MW)	Q (MVar)	Lagging p.f (pu)	P (MW)	Q (MVar)	Leading p.f. (pu)		
J1007	200	±65.737	2 x 14 MVar Cap	197.6	93.2	0.9044	196.8	110.6	-0.8718	Yes	
J1020	250	±82.171	2 x 17 MVar Cap	247.0	102.9	0.9231	245.3	149.0	-0.8547	Yes	
J1030	153.5	±50.453	2 x 16 MVar Cap	150.6	56.1	0.9371	147.3	124.0	-0.7650	Yes	
J1060	500	±242.161	2 x 6 MVar Cap	488.9	210.0	0.9188	485.8	287.5	-0.8606	Yes	
J1076	200	±96.864	N/A	197.3	66.7	0.9473	195.8	147.7	-0.7983	Yes	
J1125	135	±101.25	N/A	133.1	73.0	0.8768	130.3	171.3	-0.6054	Yes	
J1142	20	±11.222	2 x 4.8 MVar Cap	19.6	17.7	0.7422	19.4	18.7	-0.7200	Yes	
J1144	96	±72	1 x 5 MVar Cap	91.9	61.4	0.8315	88.4	102.3	-0.6538	Yes	
J1147	100	±75	N/A	97.7	57.6	0.8614	96.0	106.3	-0.6702	Yes	
J1155	150	±72.6	2 x 24 MVar Cap	147.6	102.0	0.8227	147.1	87.1	-0.8605	Yes	
J1158	70	±33.9	2 x 11 MVar Cap	68.9	43.8	0.8439	68.2	56.7	-0.7690	Yes	
J1184	50	±24.527	2 x 7 MVar Cap	49.3	30.3	0.8520	48.9	41.5	-0.7624	Yes	
J962	100	±32.868	2 x 13.5 MVar Cap	98.9	45.1	0.9099	98.6	38.5	-0.9315	Yes	

3. Model Development and Study Assumptions

3.1. Base Case Models

The origin of the DPP 2018 April South models is the MTEP18 models with the Bench Cases including all pre-queued projects and associated Network Upgrades, while the Study Cases contain all of the interconnection requests in DPP 2018 April South Phase II, in addition to all the facilities in the Bench Cases.

- Bench Cases
 - APR18-PhaseII-2023SUM-Bench
- Study Cases
 - APR18-PhaseII-2023SUM-Study

3.2. Monitored Elements

Under NERC category P0 conditions (system intact) branches were monitored for loading above the normal rating (PSS®E Rating A), and for NERC category P1-P7 conditions branches were monitored for emergency rating (PSS®E Rating B). Voltage limits were specified for system intact and contingent conditions as per applicable Transmission Owner Planning Criteria.

3.3. Contingencies

The following contingencies were considered in the steady state analysis:

- 1) NERC Category P0 (system intact -- no contingencies)
- 2) NERC Category P1 contingencies
 - a. Single element outages, at buses with a nominal voltage of 68 kV and above
 - b. Multiple element NERC Category P1 contingencies
- 3) NERC Category P2-P7 contingencies
- 4) For all the contingencies and post-disturbance analyses, cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment disabled (fixed) and switched shunt adjustment enabled.

3.4. Study Methodology

Non-linear (AC) contingency analysis was performed on the benchmark and study cases, and the incremental impact of the DPP 2018 April South generating facilities was evaluated by comparing the steady state performance of the transmission system in the Bench and Study Cases. Analyses used PSS®E version 33.11.0 and TARA version 1901.

3.5. Performance Criteria

A branch is considered a thermal constraint if the following conditions are met:

- 1) The generator has a larger than twenty percent (20%) sensitivity factor on the overloaded facilities under post-contingent condition (see NERC TPL) or five percent (5%) sensitivity factor under system-intact condition, or
- 2) The overloaded facility or the overload-causing contingency is at generator's outlet, or
- 3) The megawatt impact due to the generator is greater than or equal to twenty percent (20%) of the applicable rating (normal or emergency) of the overloaded facility, or
- 4) For any other constrained facility, where none of the Study Generators meet one of the above criteria, however, the cumulative MW impact of the group of study generators is greater than twenty percent (20%) of the rating of the facility, then only those study generators whose individual MW impact is greater than five percent (5%) of the rating of the facility and has a distribution factor (DFAX) greater than five percent (5%) will be responsible for mitigating the cumulative MW impact constraint, or
- 5) Impacts on Affected Systems would be classified as injection constraints based on the Affected Systems' criteria, or
- 6) Any other applicable Transmission Owner Local Planning Criteria are met.

A bus is considered a voltage constraint if both of the following conditions are met:

- 1) The bus voltage is outside of the applicable normal or emergency limits for the post change case, and
- 2) The change in bus voltage is greater than 0.01 per unit

All generators must mitigate thermal injection constraints and voltage constraints in order to obtain any type of Interconnection Service. Further, all generators requesting NRIS must mitigate constraints found by using the Deliverability algorithm to meet the system performance criteria for NERC category P1 events, if DFAX due to the study generator is equal to or greater than 5%.

4. Thermal Analysis

The thermal analysis results for 2023 Summer show generator projects J1020 and J1144 causing constraints. The details pertaining to the thermal analysis can be found in Appendix C – MISO ERIS Analysis (CEII).



4.1. J1020

Thermal analysis determined that this generator contributes to the following constraints:

1. Missionary 161/69 kV autotransformer #1

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to replace the Missionary 161/69 kV autotransformer #1. The cost estimate is \$1.41 million, with the project allocated 100% of the cost.

2. Missionary 161/69 kV autotransformer #2

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to replace the Missionary 161/69 kV autotransformer #2. The cost estimate is \$1.41 million, with the project allocated 100% of the cost.

4.2. J1144

Thermal analysis determined that this generator contributes to the following constraints:

1. J1144 POI – Columbia 161 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade the J1144 POI to Columbia 161 kV line. The cost estimate is \$0.6 million, with the project allocated 100% of the cost.

5. Voltage Analysis

The voltage analysis results for 2023 Summer show generator projects J1007, J1060, J1076, J1125, J1142, J1146, J1147, J1155, J1158, J1184, and J962, causing voltage violations. Several voltage constraints are mitigated by MTEP Project 9823, Target Appendix A in MTEP19, therefore no mitigation is required and no study project is assigned any cost for the mitigation of said constraints. The details pertaining to the voltage analysis can be found in Appendix C – MISO ERIS Analysis (CEII).

5.1. J1007

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 20 MVar capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 6.8% of the cost.

2. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 20 MVar capacitor bank at the Danville 115 kV substation. The cost estimate is \$2.5 million, with the project allocated 6.1% of the cost.

3. Worst Voltage at Plantation Pipeline 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of two 20 MVar capacitor banks at the Fernwood 115 kV substation. The cost estimate is \$6.0 million, with the project allocated 22.7% of the cost.



4. Worst Voltage at Franklin 500 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of three 43.2 MVar capacitor banks at the Franklin 115 kV substation. The cost estimate is \$10.0 million, with the project allocated 20.0% of the cost.

5.2. J1060

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 20 MVar capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 16.7% of the cost.

2. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 20 MVar capacitor bank at the Danville 115 kV substation. The cost estimate is \$2.5 million, with the project allocated 14.4% of the cost.

3. Worst Voltage at Plantation Pipeline 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of two 20 MVar capacitor banks at the Fernwood 115 kV substation. The cost estimate is \$6.0 million, with the project allocated 37.9% of the cost.

4. Worst Voltage at Franklin 500 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of three 43.2 MVar capacitor banks at the Franklin 115 kV substation. The cost estimate is \$10.0 million, with the project allocated 20.0% of the cost.

5.3. J1076

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 20 MVar capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 43.9% of the cost.

2. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 20 MVar capacitor bank at the Danville 115 kV substation. The cost estimate is \$2.5 million, with the project allocated 44.7% of the cost.

5.4. J1125

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed



constraint. The planned upgrade is the addition of a 20 MVAR capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 2.3% of the cost.

2. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 20 MVAR capacitor bank at the Danville 115 kV substation. The cost estimate is \$2.5 million, with the project allocated 2.3% of the cost.

3. Worst Voltage at Plantation Pipeline 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of two 20 MVAR capacitor banks at the Fernwood 115 kV substation. The cost estimate is \$6.0 million, with the project allocated 3.0% of the cost.

5.5. J1142

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 20 MVAR capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 2.3% of the cost.

2. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 20 MVAR capacitor bank at the Danville 115 kV substation. The cost estimate is \$2.5 million, with the project allocated 3.0% of the cost.

5.6. J1146

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Plantation Pipeline 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of two 20 MVAR capacitor banks at the Fernwood 115 kV substation. The cost estimate is \$6.0 million, with the project allocated 1.5% of the cost.

5.7. J1147

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 20 MVAR capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 0.8% of the cost.

2. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 20 MVAR capacitor bank at the Danville 115 kV substation. The cost estimate is \$2.5 million, with the project allocated 1.5% of the cost.



5.8. J1155

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 20 MVAR capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 3.8% of the cost.

2. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 20 MVAR capacitor bank at the Danville 115 kV substation. The cost estimate is \$2.5 million, with the project allocated 3.0% of the cost.

3. Worst Voltage at Plantation Pipeline 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of two 20 MVAR capacitor banks at the Fernwood 115 kV substation. The cost estimate is \$6.0 million, with the project allocated 13.6% of the cost.

4. Worst Voltage at Franklin 500 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of three 43.2 MVAR capacitor banks at the Franklin 115 kV substation. The cost estimate is \$10.0 million, with the project allocated 13.3% of the cost.

5.9. J1158

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 20 MVAR capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 11.4% of the cost.

2. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 20 MVAR capacitor bank at the Danville 115 kV substation. The cost estimate is \$2.5 million, with the project allocated 12.1% of the cost.

5.10. J1184

Voltage analysis determined that this generator contributes to the following constraints:

3. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 20 MVAR capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 9.8% of the cost.

4. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed



constraint. The planned upgrade is the addition of a 20 MVAR capacitor bank at the Danville 115 kV substation. The cost estimate is \$2.5 million, with the project allocated 11.4% of the cost.

5.11. J962

Voltage analysis determined that this generator contributes to the following constraints:

1. Worst Voltage at Mt. Olive 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 20 MVAR capacitor bank at the Mt. Olive 230 kV substation. The cost estimate is \$2.5 million, with the project allocated 2.3% of the cost.

2. Worst Voltage at Grambling 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of a 20 MVAR capacitor bank at the Danville 115 kV substation. The cost estimate is \$2.5 million, with the project allocated 1.5% of the cost.

3. Worst Voltage at Plantation Pipeline 115 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of two 20 MVAR capacitor banks at the Fernwood 115 kV substation. The cost estimate is \$6.0 million, with the project allocated 21.2% of the cost.

4. Worst Voltage at Franklin 500 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is the addition of three 43.2 MVAR capacitor banks at the Franklin 115 kV substation. The cost estimate is \$10.0 million, with the project allocated 46.7% of the cost.

6. Stability Analysis

Stability analysis shows that the study projects did not adversely impact the system. An additional stability study capturing the Entergy Local Planning Criteria (LPC) for new generation interconnections was also performed by MISO and shows that study projects did not adversely impact the system under the Entergy LPC for new generation interconnections. The details pertaining to the stability analysis can be found in Appendix D – MISO Stability Analysis (CEII) and Appendix E – Entergy Local Planning Criteria Stability Analysis (CEII).

6.1. Model Development

The following summer peak 2023 models were developed based on Phase II study models. The Entergy LPC stability models were developed based on the Phase II stability study models and were adjusted in order to comply with Entergy's LPC by fully dispatching nearby local generation.

- Bench Case
 - APR18-PhaseII-2023SUM-Bench
 - APR18-PhaseII-2023SUM-Bench_Entergy LPC
- Study Case
 - APR18-PhaseII-2023SUM-Study
 - APR18-PhaseII-2023SUM-Study_Entergy LPC

6.2. Study Methodology

The purpose of the study is to identify potential angular instabilities, voltage dip violations, and damping violations, if any, due to the interconnection of the projects in the DPP 2018 April South study cycle under disturbance conditions, and the impact of all study projects on the system stability performance.



The fault scenarios simulated in this study cover faults simulated as part of the MTEP18 analysis as well as selected three-phase (3PH) faults with normal clearing and single line to ground (SLG) faults with delayed clearing. Dynamic simulations of fault scenarios were performed using the DSATools TSAT program (version 18.0.10).

Fault scenarios were first simulated using the study case and the results were reviewed. For scenarios that exhibited instability, the bench case was simulated such that the stability performance with and without the proposed interconnection projects could be compared. Any new stability problems attributed to the proposed interconnection projects are flagged and reported.

For each fault, rotor angles, speed deviation, and electrical power outputs of the study generators and the generators in the proximity were monitored. Voltages at selected buses, including all POI buses of the study projects and all future buses, were also monitored.

In addition Section 10 of the “Entergy Transmission Local Planning Criteria”² states the following:

- **New Generator Interconnections:** *For Generator Interconnection (GI) Stability Studies and TPL Reliability Stability Studies only, new generation requests (for GI studies) or generation with an executed Generator Interconnection Agreement (GIA) since performance of the last reliability assessment (for TPL studies) will be studied with all generation local to the new generator at full output to determine whether the aggregate of the generation can be injected into the transmission system without experiencing transient stability issues for local faults.*

For the purposes of the Entergy LPC study local generation has been defined as generation located within one breaker substation from the Point of Interconnection if interconnecting to an existing substation or located within one breaker substation from a termination substation when tapping a line. A summary of the generation dispatch for the Entergy LPC analysis has been tabulated in Appendix E – Entergy Local Planning Criteria Stability Analysis (CEII).

The fault scenarios simulated in the Entergy LPC study cover selected 3PH and 3PH P3 contingency faults with normal clearing. In each fault scenario, a generator is disconnected at 0.5 seconds and the simulation continues to run until 5.0 seconds at which point the fault is initiated and the total simulation run time is 20.0 seconds.

6.3. Study Criteria

Transient stability study criteria are based upon Section 11.A of the “Entergy Transmission Local Planning Criteria”.

In addition, all renewable study projects are subject to the voltage ride-through and frequency ride-through criteria specified in NERC PRC-024-2 (“Generator Frequency and Voltage Protective Relay Settings”³) to check if the projects remain connected during frequency and voltage excursions. Specifically, PRC-024-2 mandates that protective relaying should be set in such a way that:

- Voltage Ride-Through: a generator shall withstand zero voltage at the POI (typically the primary side of the station transformer) for up to 0.15 seconds (9 cycles) and the ensuing voltage recovery period for three phase faults.
- Frequency Ride-Through: a generator shall maintain continuous operation between 59.5 and 60.5 Hz.

² <https://cdn.misoenergy.org/Entergy%20TO%20Planning%20Criteria108226.pdf>

³ <http://www.nerc.com/pa/Stand/Reliability%20Standards/PRC-024-2.pdf>



6.4. Study Results

The stability analyses showed that the study projects did not adversely impact the system. The complete list of 3PH and SLG faults simulated as well as their corresponding results and plots are included in Appendix D – MISO Stability Analysis (CEII) and Appendix E – Entergy Local Planning Criteria Stability Analysis (CEII).

7. Short Circuit Analysis

Short circuit analysis was performed utilizing ASPEN software. Single line to ground faults and three phase faults were evaluated for pre- and post-project cases (similar to Bench and Study models). The short circuit studies for the DPP 2018 April South study cycle indicate that the study generators do not have adverse impacts on circuit breaker capability. The details pertaining to the short circuit analysis are presented in Appendix F – Short Circuit Study Reports (CEII).

8. Affected System Impact Study

The details pertaining to the AECl, MLGW, PowerSouth, SOCO, and TVA Affected Systems studies are in Appendix G – AECl Affected Systems Study Report (CEII), Appendix H – MLGW Affected Systems Study Report (CEII), Appendix I – PowerSouth Affected Systems Study Report (CEII), Appendix J – SOCO Affected Systems Study Report (CEII), Appendix L – TVA Affected Systems Study Report (CEII), and Appendix M – SPP Affected Systems Report (CEII).

8.1. J1007

The AECl Study identified that this generator contributes to the following constraints:

1. Palmyra 345/161 kV transformer

Per AECl cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to add a second 345/161 kV transformer at Palmyra with a cost estimate of \$9.1 million. The project is allocated 1.32% of the cost.

2. Auxvasse – Kingdom City 69 kV line

Per AECl cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to do a rebuild of the Auxvasse – Kingdom City 69 kV line with a cost estimate of \$3.6 million. The project is allocated 3.51% of the cost.

8.2. J1020

The SOCO Study identified that this generator contributes to the following constraints:

1. Newton – Sweatt 115 kV line

Per SOCO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to add a new South Bay Springs to SMEPA 161/115 kV intertie with a cost estimate of \$15.0 million. The project is allocated 100% of the cost.

2. Lumberton – Carriere SW 115 kV line

Per SOCO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to add reconductor approximately 13 miles of 115 kV transmission line with a cost estimate of \$13.0 million. The project is allocated 100% of the cost.

3. Lucedale – Watson 115 kV line

Per SOCO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the



listed constraint. The planned upgrade is to add upgrade approximately 18 miles of 115 kV transmission line with a cost estimate of \$1.08 million. The project is allocated 100% of the cost.

4. Watson – Ocean Springs 230 kV line

Per SOCO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to add reconductor approximately 18 miles of 230 kV transmission line with a cost estimate of \$22.0 million. The project is allocated 100% of the cost.

5. NW D'Iberville – Wade 230 kV line

Per SOCO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to add reconductor approximately 24 miles of 230 kV transmission line with a cost estimate of \$24.0 million. The project is allocated 100% of the cost.

6. Barry – Wade 230 kV line

Per SOCO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to loop in at Ellicott 230 kV station with a cost estimate of \$2.0 million. The project is allocated 100% of the cost.

8.3. J1030

No affected systems mitigations were found to be required for this generator.

8.4. J1060

The AECl Study identified that this generator contributes to the following constraints:

1. Palmyra 345/161 kV transformer

Per AECl cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to add a second 345/161 kV transformer at Palmyra with a cost estimate of \$9.1 million. The project is allocated 3.32% of the cost.

2. Auxvasse – Kingdom City 69 kV line

Per AECl cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to do a rebuild of the Auxvasse – Kingdom City 69 kV line with a cost estimate of \$3.6 million. The project is allocated 8.42% of the cost.

The MLGW Study identified that this generator contributes to the following constraints:

1. Shady Grove – N.E. Gate 161 kV

Per MLGW cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to uprate the Shady Grove – N.E. Gate 161 kV line with a cost estimate of \$300,000. The project is allocated 100% of the cost.

The TVA Study identified that this generator contributes to the following constraints:

1. Freeport – Horn Lake 161 kV tie line



Per TVA cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to uprate 3.5 miles of the line with a cost estimate of \$800,000. The project is allocated 80% of the cost.

2. Freeport 500/161 kV transformer

Per TVA cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to replace the 161 kV jumpers for the transformer with a cost estimate of \$100,000. The project is allocated 100% of the cost.

8.5. J1076

No affected systems mitigations were found to be required for this generator.

8.6. J1125

The AECl Study identified that this generator contributes to the following constraints:

1. Palmyra 345/161 kV transformer

Per AECl cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to add a second 345/161 kV transformer at Palmyra with a cost estimate of \$9.1 million. The project is allocated 0.89% of the cost.

2. Auxvasse – Kingdom City 69 kV line

Per AECl cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to do a rebuild of the Auxvasse – Kingdom City 69 kV line with a cost estimate of \$3.6 million. The project is allocated 2.39% of the cost.

8.7. J1142

No affected systems mitigations were found to be required for this generator.

8.8. J1144

No affected systems mitigations were found to be required for this generator.

8.9. J1146

No affected systems mitigations were found to be required for this generator.

8.10. J1147

No affected systems mitigations were found to be required for this generator.

8.11. J1155

No affected systems mitigations were found to be required for this generator.

8.12. J1158

No affected systems mitigations were found to be required for this generator.



8.13. J1184

No affected systems mitigations were found to be required for this generator.

8.14. J962

The TVA Study identified that this generator contributes to the following constraints:

1. Freeport – Horn Lake 161 kV tie line

Per TVA cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to uprate 3.5 miles of the line with a cost estimate of \$800,000. The project is allocated 20% of the cost.

9. Deliverability Analysis

9.1. Introduction

Generator interconnection projects have to pass Generator Deliverability Study to be granted NRIS. If the generator is deemed not fully deliverable, the customer can choose either to change the project to an Energy Resource project or to proceed with the system upgrades that will make the generator fully deliverable. Generator Deliverability Study ensures that the Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak condition without getting bottled up. The wind and solar generators are tested at 100% of their maximum output level which then can be used to meet Resource Adequacy obligations, under Module E, of the MISO Transmission and Energy Market Tariff (TEMT).

The MISO Generator Deliverability Study whitepaper describing the algorithm can be found in BPM 015 – Generation Interconnection, Appendix C.

9.2. Determining the MW Restriction

If one facility is overloaded based on the assessed “severe yet credible dispatch” scenario described in the study methodology, and the generator under study has a DF greater than 5%, part or all of its output is not deliverable. The restricted MW is calculated as following:

$$(\text{MW restricted}) = (\text{worst loading} - \text{MW rating}) / (\text{generator sensitivity factor})$$

If the result is larger than the maximum output of the generator, 100% of this generator's output is not deliverable.

9.3. Deliverability Study Results

The limiting constraints (monitored facilities – contingency facility pairs) seen in the deliverability analysis for the 2023 Summer case are summarized in Appendix N – MISO Deliverability Analysis (CEII).

9.3.1. J1007

This generator is determined to be deliverable up to 182.4 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 6 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 6: NRIS Results for J1007

J1007 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERS and case assumptions)		182.4 MW (91.18%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Franklin – McKnight 500 kV	200.0	5.02%	J1007, J1060, J1125	22,267,419	94,000,000

9.3.2. J1020

This generator is found to be fully deliverable up to 250 MW without any Network Upgrades

9.3.3. J1060

This generator is determined to be deliverable up to 455.9 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 7 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 7: NRIS Results for J1060

J1060 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERS and case assumptions)		455.9 MW (91.18%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Franklin – McKnight 500 kV	500.0	5.07%	J1007, J1060, J1125	56,223,014	94,000,000

9.3.4. J1076

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 8 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 8: NRIS Results for J1076

J1076 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERS and case assumptions)		0.0 MW (0.0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Tezcuco – Frisco 230 kV #1	0.0	12.48%	J1076, J1142, J1158	863,806	2,500,000
Tezcuco – Frisco 230 kV #2	200.0	12.43%	J1076, J1142, J1158	860,332	

9.3.5. J1125

This generator is determined to be deliverable up to 123.1 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 9 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 9: NRIS Results for J1125

J1060 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERIS and case assumptions)		123.1 MW (91.18%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Franklin – McKnight 500 kV	135.0	5.18%	J1007, J1060, J1125	15,509,568	94,000,000

9.3.6. J1142

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 10 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 10: NRIS Results for J1142

J1142 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERIS and case assumptions)		0.0 MW (0.0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Tezcuco – Frisco 230 kV #1	0.0	12.48%	J1076, J1142, J1158	86,381	2,500,000
Tezcuco – Frisco 230 kV #2	200.0	12.43%	J1076, J1142, J1158	86,033	

9.3.7. J1144

This generator is found to be fully deliverable up to 96 MW without any Network Upgrades

9.3.8. J1146

This generator is found to be fully deliverable up to 10 MW without any Network Upgrades

9.3.9. J1155

This generator is found to be fully deliverable up to 150 MW without any Network Upgrades.

9.3.10. J1158

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 11 shows the NRIS results and cost estimates determined in the NRIS analysis.

Table 11: NRIS Results for J1158

J1158 Deliverable (NRIS) Amount in 2023 Case: (Conditional on ERIS and case assumptions)		0.0 MW (0.0%)			
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Projects Associated with NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total Cost of Upgrade (\$)
Tezcuco – Frisco 230 kV #1	0.0	12.48%	J1076, J1142, J1158	302,332	2,500,000
Tezcuco – Frisco 230 kV #2	200.0	12.43%	J1076, J1142, J1158	301,116	

9.3.11. J1184

This generator is found to be fully deliverable up to 50 MW without any Network Upgrades.

10. Shared Network Upgrades Analysis

Shared Network Upgrade (SNU) Analysis, which tests for Network Upgrades driven by higher queued interconnection projects, was performed for this System Impact Study.

The maximum MW impacts and SNU cost allocations appear in Table 12.

Table 12: Maximum MW Impact and SNU Cost Allocations

Network Upgrades	Project Study Cycle	Projects sharing cost	MW Contribution	Total NU Cost (\$)	Cost Responsibility (\$)
Kuhn Rd – Ebony 161 kV Rebuild	DPP-2017-AUG	J934	133.02	4,900,000	3,508,251
	DPP-2018-APR	J1007	16.30		429,894
	DPP-2018-APR	J1060	13.29		350,509
	DPP-2018-APR	J1125	23.18		611,346

11. Cost Allocation

The cost allocation of Network Upgrades (NU) for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of NRIS as of the draft System Impact Study report date.

11.1. Cost Assumptions for Network Upgrades

The cost estimate for each NU identified in the System Impact Study is provided by the corresponding Transmission Owner.

11.2. Cost Allocation Methodology

The costs of NU for a set of generation projects (one or more sub-groups or entire group with identified NU) are allocated based on the MW impact from each project on the constrained facilities in the Study Case.

Cost Allocation Methodology for Thermal Constraints

1. With all study group generation projects dispatched in the Study Case, all thermal constraints are identified.
2. Distribution factor from each project on each constraint is obtained.
3. For each thermal constraint, the maximum MW contribution (increasing flow) from each project is then calculated in the Post Case without any Network Upgrades.
4. For each thermal constraint, the cost estimates for one or a subset of NU are provided by the corresponding Transmission Owner.
5. The cost of each NU is allocated based on the pro rata share of the MW contribution from each project on the constraints mitigated or partly mitigated by this NU. The methodology to determine the cost allocation of one NU is:

$$\text{Project A Cost Portion of NU} = \frac{\text{Max(Proj. A MW contribution on constraint)}}{\sum_i \text{Max(Proj. i MW contrution on constraint)}}$$

6. The total NU costs for each project are calculated if more than one NU is required.



Appendix A – Cost Allocation Summary (CEII)

Appendix B – Network Upgrade Table Per Project (CEII)

Appendix C – MISO ERIS Analysis (CEII)

Appendix D – MISO Stability Analysis (CEII)

Appendix E – Entergy Local Planning Criteria Stability Analysis (CEII)

Appendix F – Short Circuit Study Reports (CEII)

Appendix G – AECI Affected Systems Study Report (CEII)

Appendix H – MLGW Affected Systems Study Report (CEII)

Appendix I – PowerSouth Affected Systems Study Report (CEII)

Appendix J – SOCO Affected Systems Study Report (CEII)

Appendix L – TVA Affected Systems Study Report (CEII)

Appendix M – SPP Affected Systems Study Report (CEII)

Appendix N – MISO Deliverability Analysis (CEII)